

Answer the following

1 x 20 = 20 Marks

19. (a) Diagonalize the matrix $A = \begin{bmatrix} 2 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{bmatrix}$ by means of orthogonal transformation. (12)

(b) Establish the following relationship between Beta and Gamma integrals: $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$. (8)

UNIVERSITY
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End Semester Examinations

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Course Code: MAT101R01

Course: ENGINEERING MATHEMATICS - I

Question Paper No.: U0545

Duration: 3 hours

Max. Marks: 100

PART - A

Answer all the questions

10 x 2 = 20 Marks

1. If $\tanh(x/2) = \tan(\theta/2)$, Find the value of x .
2. If $\tan \log(x + iy) = a + ib$, where $a^2 + b^2 \neq 1$, prove that $\tan \log(x^2 + y^2) = \frac{2a}{1 - a^2 - b^2}$.
3. Check whether the Vectors $X = (1, 1, 2)$, $Y = (1, 2, 5)$ and $Z = (5, 3, 4)$ are linearly dependent or independent.
4. For what values of 'k' the following system of equations $kx + y + z = 1$; $x + ky + z = 1$; $x + y + kz = 1$ has a unique solution.
5. The product of two eigenvalues of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ is 16. Find the third eigen value.
6. Find the index and signature of the Q.F. $x^2 + 2y^2 - 3z^2$.
7. For what value of k, the equation $2x^3 - 9x^2 + 12x + k = 0$ has a double root.

10. Evaluate $\int_0^1 \int_0^2 \int_0^3 xyz \, dx \, dy \, dz$.

PART - B

Answer all the questions

4 x 15 = 60 Marks

11. (a) Expand $\sin^3 \theta \cos^5 \theta$ in a series of sines of multiples of θ . (8)

(b) Prove that $\tanh^{-1} x + \tanh^{-1} y = \tanh^{-1}(x + y / (1 + xy))$. (7)

(OR)

12. (a) Separate the real and imaginary parts $(\sqrt{i})^{\sqrt{i}}$. (8)

(b) If $\tan(\theta + i\phi) = \tan \beta + i \sec \beta$, prove that $e^{2\phi} = \pm \cot(\beta/2)$. (7)

13. (a) Discuss the consistency of the system and if consistent, solve the equations

$$4x - 2y + 6z = 8$$

$$x + y - 3z = -1$$

$$15x - 3y + 9z = 21.$$

(5)

(b) Express $A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$ as singular value decomposition. (10)

(OR)

14. (a) Find the reflection of the point (3,2) about the line $2x + 4y + 7 = 0$. (5)

vi) Adj. A.

15. (a) Solve the equation $27x^3 + 42x^2 - 28x - 8 = 0$ whose roots are in Geometric Progression.

(b) If a, b, c are the roots of $8x^3 - 4x^2 + 6x - 1 = 0$. Find the equation whose roots are $a + \frac{1}{2}$, $b + \frac{1}{2}$, $c + \frac{1}{2}$.

(OR)

16. (a) Determine all the roots of the equation $6x^6 - 25x^5 + 31x^4 - 31x^2 + 25x - 6 = 0$.

(b) Determine the value of the following series

$$\frac{4}{8} + \frac{4.12}{18.27} + \frac{4.12.20}{18.27.36} + \dots$$

17. (a) Compute the area of astroid $x^{2/3} + y^{2/3} = a^{2/3}$, using parametric functions.

(b) Determine the value of the integral $\int_0^{\pi/2} \sqrt{\cot \theta} \, d\theta$.

(OR)

18. (a) Change the order of integration in $\int_0^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} dy \, dx$ and then evaluate it.

(b) Compute the value of the following integral $\int_1^e \int_1^{\log y} \int_1^{e^x} \log z \, dx \, dy \, dz$.

